

Lingli Kong, PhD

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SUMMARY

PhD computational scientist focused on cryo-EM/structural analytics and scientific software (C++/Python, PyTorch). Built production-grade algorithms in *cisTEM* and deep-learning classifiers that improved detection/SNR on EM datasets. Partner with discovery scientists to deliver in silico workflows and communicate results clearly. Interested in applying ML/generative methods to biologics design; quick to ramp on Rosetta/Schrödinger as part of end-to-end structural workflows.

SKILLS

Languages: C++, Fortran, Python

ML/CV: PyTorch, SciPy, scikit-learn, OpenCV, CuPy, LSTM, CNN

Other: Linux, MATLAB, PyQt, Flask, Docker, HPC, CUDA

RESEARCH EXPERIENCE

University of Massachusetts Chan Medical School

Worcester, MA

Research Specialist

06/2022 – Present

Membrane Protein Detection From Liposomes Using 2D Template Matching (2DTM) and Deep Learning

- Developed a deep-learning classifier using PyTorch (dual-head CNN encoder) that can overcome the hard threshold problem in 2DTM.
- Improved the true positive detection rate for membrane proteins (around 300 kDa) in thick ice (around 200 nm), from 38 % to 92 % for single particle samples, and from 20 % to 58 % for samples where the protein is embedded in membrane.
- Generalized the refined method and the deep-learning classifier to experimental liposome micrographs, and eliminated the degree of template bias by 32 %.

Distortion Modeling and Correction for Cryo-EM Movie Frames

- Developed Unbend program in *cisTEM* software (C++) by implementing a 3D spline model to correct beam-induced motions at the pixel level in cryo-EM movie frames.
- Integrated a local motion visualization panel into the *cisTEM* graphical user interface (GUI).
- Evaluated the performance using 2DTM method, and observed signal-to-noise ratio (SNR) improvement of 3 to 8 % and number of detected particles increase of up to 300 % across different types of samples.

Tilt Estimation and Absolute Pre-Tilt Determination Using CTF Information

- Implemented a tilt estimation method based on local defocus variation across the cryo-EM micrographs in C++ to the CTFFIND5 program.
- Proposed an algorithm for determining the absolute sample pre-tilt by fitting the rotation matrix using downhill-simplex method, and coded a corresponding program, `fit_tilt_model`, in C++ for *cisTEM*.

University of California, Irvine

Irvine, CA

Assistant Specialist

02/2022 – 06/2022

Software Development for Scanning Transmission Electron Tomography Processing

- Developed Python software with a GUI to automate tomography alignment and process STEM images.

Deep-Learning Method Development for Electron Energy Loss Spectroscopy (EELS) Spectra Processing

- Implemented a CNN-BiLSTM network to automate the core-loss edges recognition on EELS spectra, achieving an accuracy of 94.9%.

Lawrence Berkeley National Laboratory

Berkeley, CA

Postdoctoral Scholar

01/2020 – 01/2022

Analysis of HIV Infection with Liquid-Cell TEM Technology

- Analyzed stages of HIV infection in HeLa cells based on the 3D reconstructions from individual particle electron tomography (IPET) technique and missing wedge algorithm.

OTHER COMPUTATIONAL EXPERIENCE

University of Wyoming

Laramie, WY

Graduate Research Assistant

09/2014 – 08/2019

Monte Carlo (MC) Simulation Code Development and Mathematical Modeling for Gas Adsorption

- Programmed Grand Canonical Monte Carlo (GCMC) simulation for gas adsorption in porous materials from scratch in Fortran.
- Derived a closed-form adsorption model for slit-pores, which works from micro to macro scales and well captures the effects of bulk conditions and properties of the adsorbate.
- Proposed an adsorption model based on statistical thermodynamics that can predict all types of adsorption isotherms (classified by IUPAC report) in various types of porous materials.

EDUCATION

Ph.D. in Chemical Engineering

08/2019

University of Wyoming

Laramie, WY

B.S. in Petroleum Engineering

06/2014

China University of Petroleum

Qingdao, P.R. China

SELECTED PUBLICATIONS

- **Kong, L.**, Zottig, X., Elferich, J., & Grigorieff, N. (2025). Unbend: Correction of local beam-induced sample motion in cryo-EM images using a 3D spline model. *eLife* (under review).
- **Kong, L.**, Cossio, P., Hanson, S., Furukawa, H., & Grigorieff, N. (2025). Membrane protein detection from liposomes in cryo-EM images using 2D template matching and deep learning. (submitted to Machine Learning in Structural Biology)
- Elferich, J., Kaminek, M., **Kong, L.**, Odriozola, A., Kukulski, W., Zuber, B., & Grigorieff, N. (2025). In situ high-resolution cryo-EM reconstructions from CEMOVIS. *IUCr*, 23, 502–510.
- Elferich, J., **Kong, L.**, Zottig, X., & Grigorieff, N. (2024). CTFFIND5 provides improved insight into quality, tilt and thickness of TEM samples. *eLife*, 97227.1.
- **Kong, L.**, Liu, J., Zhang, M., Lu, Z., Xue, H., Ren, A., Liu, J., Li, J., Ling, W. L., & Ren, G. (2023). Facile hermetic TEM grid preparation for molecular imaging of hydrated biological samples at room temperature. *Nature Communications*, 14, 5641.
- **Kong, L.**, Ji, Z., & Xin, H. (2022). Electron energy loss spectroscopy database synthesis and automation of core-loss edge recognition by deep-learning neural networks. *Scientific Reports*, 12, 22183.
- **Kong, L.**, & Adidharma, H. (2019). A new adsorption model based on generalized van der Waals partition function for the description of all types of adsorption isotherms. *Chemical Engineering Journal*, 375, 122112.
- **Kong, L.**, & Adidharma, H. (2019). A generalized van der Waals model for light gases adsorption prediction in IRMOFs. *Physical Chemistry Chemical Physics*, 21, 8906–8914.
- Rony, A. H., **Kong, L.**, Lu, W., Dejam, M., Adidharma, H., Gasem, K. A. M., Zheng, Y., Norton, U., & Fan, M. (2019). Kinetics, thermodynamics, and physical characterizations of corn stover (*Zea mays*) for solar biomass pyrolysis potential analysis. *Bioresource Technology*, 284, 466–473.
- **Kong, L.**, & Adidharma, H. (2018). Adsorption of simple square-well fluids in slit nanopores: Modeling based on generalized van der Waals partition function and Monte Carlo simulation. *Chemical Engineering Science*, 177, 323–332.